

1. Scenario: Spelling Correction in a Text Editor

Question: You're creating a simple text editor. How would you correct spelling mistakes in a sentence?

- Tokenize the sentence into individual words.
- Check each word against a dictionary or vocabulary.
- Flag words that are not found.
- Generate suggestions using edit distance (how many characters differ).
- Rank suggestions using word frequency and context.
- Replace the word with the most probable correction

2. Scenario: Counting Words in Customer Reviews

Question: You want to analyze customer reviews. How will you count how many times each word is used?

- Convert all text to lowercase.
- Remove punctuation and special characters.
- Tokenize the text into words.
- Remove stopwords like "the", "is", "and".
- Count how many times each word appears.
- Store results in a frequency table or dictionary.

3. Scenario: Identifying Questions in Chat Messages

Question: In a chatbot, how do you know if the user typed a question or not?

- Check if the sentence ends with a question mark (?).
- Look for question words like what, why, how, when, where.
- Analyze sentence structure using part-of-speech tagging.
- Use a classifier (ML model) trained on questions vs statements.
- Return TRUE if it matches question patterns, else FALSE.

4. Scenario: Language Detection for Multilingual Inputs

Question: A user enters text on your website, and you need to detect the language. How would you do that?

- Clean the text (remove symbols, extra spaces).
- Break the text into small letter groups (like "ho", "ola", "bon").
- Compare these patterns with known language samples.
- Use a pre-trained language detection model.
- Pick the language with the highest probability score.

5. Scenario: Creating Tags from Blog Articles

Question: You want to create automatic tags for blog articles. How can you generate useful keywords?

- Clean the text (lowercase, remove punctuation).
- Tokenize the article into words.
- Remove stopwords and very common words.
- Calculate TF-IDF (importance of each word in the article).
- Select top-ranking words as keywords.
- Group similar words using stemming or lemmatization and use that.

6. Scenario: Convert Text to Lowercase for Uniformity

Question: Why is converting all text to lowercase useful in NLP?

- “Apple”, “apple”, and “APPLE” become the same token.
- Improves matching accuracy.
- Reduces vocabulary size.
- Makes word counting and searching consistent.
- Prevents false mismatches in models.

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